

PATRICK KEOUGH

PROFESSIONAL SUMMARY

Independent LLM evaluation researcher bridging psychometric methodology and behavioral auditing of frontier AI systems. Three preprints covering sycophancy auditing, demographic bias in mental health simulation, and explanation faithfulness, totaling over 40,000 model observations across eight frontier LLMs from US and Chinese development contexts. Methodology grounded in psychometrics: continuous-severity rubrics, LLM-as-Judge pipelines validated against 236 human raters and an external cross-lineage AI judge, and non-parametric statistical inference at scale. Concurrent work shipping production AI applications across six LLM providers.

SKILLS

LLM Evaluation and Methodology: Multi-axis rubric design, LLM-as-Judge pipelines, hybrid search retrieval (Vector + BM25 FTS5), Reciprocal Rank Fusion (RRF), semantic average-linkage clustering, inter-rater reliability, explanation faithfulness.

AI-Assisted Development and Infrastructure: Build production applications using agentic CLI tooling: Claude Code, Gemini CLI, Roo-Code. Stack includes Python, FastAPI, React, Node.js, SQLite FTS5, FastMCP, FastEmbed CPU-ONNX, ChromaDB.

Statistical Methods: Non-parametric testing (Kruskal-Wallis, Mann-Whitney, Dunn's), mixed-effects models, effect-size estimation (Cohen's d, Cliff's delta), Benjamini-Hochberg FDR correction, bootstrap resampling, inter-rater agreement (Cohen's and Fleiss' kappa).

AI Tooling and APIs: OpenRouter, Claude API, Gemini API, OpenAI API, DeepSeek, GLM, FastMCP, FastEmbed, ComfyUI, Stable Diffusion. Local fine-tuning with SFT and DPO on various local models.

EXPERIENCE

Independent LLM Evaluation Researcher

Self-directed Jul 2025 to Present | Sydney, Australia

- Designed and executed three end-to-end LLM evaluation studies, totaling over 40,000 model observations across eight frontier LLMs from US and Chinese development contexts. All pipelines and datasets open-source.
- Built and validated multi-axis evaluation rubrics across five behavioral constructs (sycophancy, truthfulness, refusal specificity, faithfulness, fairwashing), achieving inter-rater agreement of Cohen's kappa = 0.78 against 236 human raters and 93.3 percent weighted agreement with an external cross-lineage AI judge.
- Documented systematic LLM failure modes invisible to standard evaluation, including a 37 percent diagnostic-threshold-crossing rate under identical-prompt re-runs, variance compression ranging from 14 to 62 percent across models, and demographic-cohort severity distortions up to Cohen's d = 1.95.
- Built production statistical pipelines using non-parametric inference (Kruskal-Wallis, Mann-Whitney, Dunn's), mixed-effects modeling, and Benjamini-Hochberg FDR correction. Local fine-tuning experience with SFT and DPO on Llama 3.1 8B.

Founder & Principal Technician

OlympicCustomPCs Aug 2023 to Mar 2025 | Seattle, WA

- Operated an independent PC building and repair business serving 30+ clients. Built custom systems to specification, diagnosed hardware and software issues, and developed Python tooling for inventory management, deal sourcing, and customer-inquiry tracking.

National Executive Policy Board Member and Chapter Policy Leader

Students for Sensible Drug Policy (volunteer) Sep 2022 to Jun 2024 | National and University of Washington Chapter

- One of two national student representatives on the executive policy board (Sep 2022 to Jun 2023). Reviewed and voted on state and federal harm-reduction policy proposals on behalf of the national student membership.
- Led the first campus Narcan distribution program in Washington state. Conducted overdose-response trainings, delivered lectures on drug policy ethics, and coordinated with the Washington State Department of Health on harm-reduction programming.

SELECTED PUBLICATIONS

The Granularity Gap: Continuous Severity Auditing of LLM Sycophancy Across Gemini Model Generations (2026). Solo author.

Continuous LLM-as-Judge framework across N = 8,830 responses from 8 Gemini variants. Documented the Alignment Tax: sycophancy-hallucination coupling nearly doubles across model generations (Fisher's $Z = 9.12$, $p < 0.001$). github.com/pskeough/The-Granularity-Gap

PsychBench: Variance Collapse and Bias Amplification in LLM Mental Health Simulation (2026).

Solo author.

N = 28,800 across 120 intersectional demographic cohorts and 4 frontier models, benchmarked against NHANES and NESARC-III. Documented SES-driven severity distortion (Cohen's $d = 1.95$) and transgender symptom suppression (Cohen's $d = -1.55$). arxiv.org/abs/2604.17359 | github.com/pskeough/PsychBench

The Explanation Gap: Faithfulness Failures in LLM Self-Reports of Demographic Bias in Mental Health Simulation (2026, in preparation). Solo author.

Four-axis LLM-as-Judge evaluation of how faithfully models explain their own documented biases. Pilot N = 522: 61.5 percent of explanations score high on fairwashing, with 21.1 percent simultaneously faithful and fairwashing. github.com/pskeough/The-Explanation-Gap

Iterated Consensus in Inductive Qualitative Coding (2026, in progress). Technical lead.

Evaluation framework extending the LLM-Quotient reliability methodology to inductive thematic analysis. Projects 43,000 open-code strings into semantic vector space, implementing two-axis reliability metrics, span stability tracking, and a six-stage verbatim recovery cascade. Private dataset.

SELECTED PROJECTS

Claude CoWrite: Local-first creative writing IDE with line-by-line LLM editing, word-level diffing, mode-separated AI roles, SSE streaming, and headless Claude CLI integration. No cloud upload. React 18, Vite, TypeScript, Node.js, Express, Claude Code CLI.

Sonic Vector: System-wide audio-mastering dashboard binding Windows Equalizer APO to active Spotify playback. Dynamic parametric filters are generated on song transitions by interpolating SAFE database crowdsourced semantic audio centroids.

Mimesis MCP: Local-first FastMCP server equipping Claude with tools to draft in a target voice. Employs a multi-stage offline RAG pipeline using BAAI/bge-small-en-v1.5 ONNX embeddings, SQLite FTS5 keyword indices, Reciprocal Rank Fusion, and real-time stylometric auditing.

EDUCATION

Bachelor of Arts, Psychology

University of Washington, Seattle *Sep 2022 to Jun 2025* | *GPA: 3.77 / 4.00* | *Dean's List: 8 of 9 quarters*

Relevant coursework: PSYCH 315 Statistics, PSYCH 209 Fundamentals of Psychological Research, JSIS B 480 Global Cybersecurity, INFO 200 Intellectual Foundations of Informatics.